Sigma Notation: Index Shifts and Convergence Conditions

Precalculus

Counting and shifting the index of a sigma sum, testing the convergence condition before using the geometric sum formula, and reading a sequence limit correctly

Directions:

- A sigma sum is defined by *both* its summand and its limits. When you shift the index, shift the limits by the same amount — the number of terms must not change.
- The infinite geometric formula $\frac{a}{1-r}$ is valid **only** when $|r| < 1$, and a must be the term the sum actually starts with. State a and r, then check the condition, *then* compute.
- On a **find the mistake** problem, name the error in words *and* show correct work. Leave answers as exact fractions.

1. Evaluate the sum.

$$\sum_{n=1}^{5}(2n+1)$$

Answer: ________________

2. How many terms does this sum have?

$$\sum_{k=4}^{19} a_k$$

Answer: ________________ terms

3. Find the mistake. Ravi wants to evaluate $\sum_{n=2}^{7}(n-1)^2$. He substitutes $m = n-1$ and writes

$$\sum_{n=2}^{7}(n-1)^2 \;=\; \sum_{m=1}^{7} m^2 \;=\; 140$$

(a) Name Ravi's error in words — what did he forget to shift? (b) Write the correctly shifted sum and evaluate it.

Answer: ____________________

4. Find the sum, or state that the series diverges.

$$\sum_{n=0}^{\infty} 3\left(\frac{2}{5}\right)^n$$

Answer: ____________________

5. Find the mistake. Tess writes

$$\sum_{n=1}^{\infty} 4\left(\frac{3}{2}\right)^n = \frac{4}{1-\frac{3}{2}} = -8$$

(a) Name the condition Tess never checked, and show that it fails here. (b) Add the first four terms to show what the partial sums are doing. (c) State what is true about this series.

Answer: ________________

6. (a) Evaluate $\lim_{n\to\infty} \frac{1}{n}$.

(b) A student writes: *"The terms go to 0, so the series $\sum \frac{1}{n}$ converges."* Explain why part (a) does not prove that claim, and name the series that shows the claim is false.

Answer: ________________

7. Find the sum, or state that the series diverges. Be careful about the value of the first term.

$$\sum_{n=2}^{\infty} \left(\frac{2}{3}\right)^n$$

Answer: ________________

8. Rewrite the sum below as an equivalent sum whose index starts at 1, then evaluate it. Show that your rewritten sum has the same number of terms as the original.

$$\sum_{n=4}^{12} (n-3)^3$$

Answer: ________________